

developed to turn plastic waste into high-performance materials, in the form of tiles and blocks. These materials are weather-resistant, acid-proof, durable, offer better structural stability at a low cost and protect the environment from plastic dumping.

This innovation is the brainchild of a team lead by Dr S.K. Dhawan, emeritus scientist, CSIR-National Physical Laboratory. The technology has been transferred to Shayna EcoUnified India Pvt Ltd in New Delhi and Addin Infra Pvt Ltd in Rajkot, with many other companies showing interest. Union minister of science and technology, Harsh Vardhan, said that this technology could serve the twin purposes of plastic waste management and better sanitation in the country.

Prashant Lingam, a distributor of these tiles to Municipal Corporation of Hyderabad, has already supplied 2323sqm (25,000-square-foot) tiles.

Challenges faced in curbing plastic waste

According to the disposal policy of the government of India, plastic waste goes to landfill sites. In the landfills, it takes more than 1000 years to completely degrade. Plastic in different states of degradation releases toxic materials that leach into the ground and pollute the groundwater.

Plastic should not be burnt, either, as burning it releases toxic gases like carbon monoxide, phosgene, nitrogen oxide and dioxin. In addition, burning one kilogram of plastic releases three kilograms of carbon dioxide—a gas that contributes to global warming. This project not only solves the problem of disposal of waste plastic bags but, also utilises these bags into tiles that can be used for designing structures.

Benefits and applications of the technology

Paras Saluja, founder and director, Shayna EcoUnified India Pvt Ltd, says



Toilet made using tiles made from waste plastic scrap; 42,000 bags were used in making a 2.1m x 1.2m x 1.2m (7-feet x 4-feet x 4-feet) structure (Credit: Dr S.K. Dhawan)

that the tiles produced by the green technology have successfully passed flammability, water absorption and mechanical strength tests, carried out as per ASTM standards.

According to Ashish Joshi, director, Addin Infra Pvt Ltd, issues like mechanical strength, flame retardancy, water permeability and UV protection from sunlight and antistatic response are the novelty of the concept.

Saluja says that the product range is vast and so are its applications. Key applications are interlocking tiles, paver tiles, interlock tiles and roof tiles. These can be used in building public toilets, shelter homes, pavements, structures and rooms for general public, providing social and economic benefits. Shayna EcoUnified strives to deploy cost-effective architectural tiles further, redirecting the recycling dimension of India towards evolution.

The role of electronics

Dr Dhawan explains, “This is a fully-automated machine setup that uses sensors and microcontrollers (MCUs) for accurate working. For example, a

hydraulic pressure machine works only when it is completely closed. Temperature, pressure, quantity and distance are monitored throughout the process using sensors, and sent to respective machines over MCUs for achieving high performance through moulding and other machines.”

Joshi says, “With a team of 15 employees the production can start. Required capital cost is ₹ 55 million for machinery. Whereas, cost of operation for the first three years is ₹ 25 million, which includes manpower, raw material and daily operational activities.”

Saluja adds, “For running the machinery, minimum 2000kVA electricity is needed.

“We only need segregated raw materials for our product. Additionally, proportional mixing of raw materials is important to keep a check on hazardous gas emissions.”

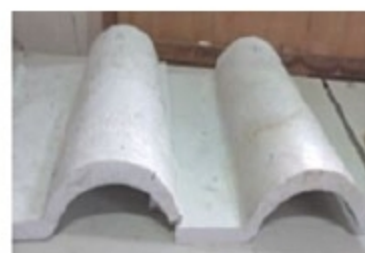
The road ahead

Since plastic consumption cannot be avoided completely, the government of India has initiated measures to recycle plastic. The market for recycled plastic products is huge. Saluja says, “Our vision is to reduce the chronic effects of plastic bags in India and build plastic-free cities and towns, along with offering long-term resolutions in terms of basic amenities like toilets and other societal structures in rural and urban areas.”

Joshi says, “Different sources of waste plastic have been identified near Surat. As of now, we have a monthly target of making a paver block that can cover 2323sqm (25,000-square-foot)—to be increased to 9290sqm (100,000-square-foot) per month in the next six months. Production of the block will consume two kilograms of plastic per square metre—total 200,000 kilograms waste plastic will be utilised in a month.

“To ensure access to sanitation facilities and development of a cleaner India, Prime Minister Narendra Modi launched Swachh Bharat Abhiyan. We have taken this mission a step ahead by introducing the concept of smart toilets made of waste plastic bags, wherein eco-friendly toilets will be constructed from these tiles.” **EFY**

—Deepshikha Shukla, technical journalist, EFY



Interlocking and roof tiles (Credit: Dr S.K. Dhawan)